

Objective:

In this project, you will design your own synthetic dataset, preprocess it, and train a Multilayer Perceptron (MLP) for a regression problem, to make predictions based on the generated data. You must not replicate the exact example covered in class—instead, explore a different relationship or pattern.

Tasks:

1. Reduce the number of datapoints
2. Introduce random noise
3. Train your previous model
4. Introduce regularization
 - Find the proper regularization for your problem, not all problems need the same layers or parameters.
5. Analysis & Report
 - Discuss and compare the results with your previous work.
 - Explain your model's performance (accuracy, loss, etc.).

Requirements:

- The real world problem must be original (not a direct copy of the class example).
- Code should be well-structured and the student should use Python, Colab/Pycharm and Keras for its development.
- Submit a brief report (3-5 pages) or give a presentation summarizing your results (2-5 minutes) (and submit the video).

Deadline: One week from today.

Last Deadline: Last day of class, maximum mark will be 1.

Rubrics:

	0	1	2
Regularization	The regularization worsened the results.	The regularization improved the results but the students needed help.	The regularization improved the results thanks to the students' own work.